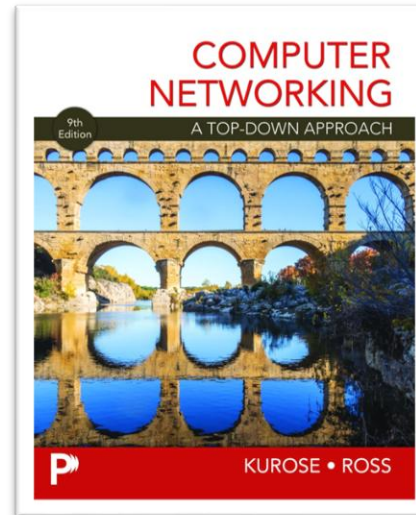


Computer Networks

Computer Science 3rd course

Bibliography

- *Computer Networking: A Top Down Approach*, 9th edition.
Jim Kurose, Keith Ross. Pearson
- Available at UC3M library
 - 8th edition also as electronic resource
 - https://www.ingebook.com/ib/NPcd/IB_Escritorio_Visualizar?cod_primaria=1000193&libro=12321



Schedule

- Course schedule

- Details the content of every class

- Generic schedule

- Available in UC3M website

- Detailed schedule

- Aula Global (**this is the good one!**)

- With date & time & lecturer & room/online per group

- All the material is accesible through Aula Global

- In the website of the ‘Magistral’ group

- The name of the resources start with letters that identify them in the schedule (si.j, L, PC, etc.)

Semana	Sesión	Grupo	DÍA	TEMÁTICA	AULA	HORA	PROFESOR
1	s1.1	M81	martes, 8 de septiembre de 2015	Introducción a las Redes de Ordenadores, Protocolos, Conmutación de Circuitos y Conmutación de Paquetes, Arquitectura de Internet, Modelo de Capas	4.1E03	11:00-13:00	Marcelo
		M83	martes, 8 de septiembre de 2015	Introducción a las Redes de Ordenadores, Protocolos, Conmutación de Circuitos y Conmutación de Paquetes, Arquitectura de Internet, Modelo de Capas	4.1E03	17:00-19:00	Marcelo
		M89	Tuesday, 8 September 2015	Introduction to Computer Networks, Protocols, Circuit Switching, Packet Switching, Internet Architecture, Protocol Layers	1.2G04	9:00-11:00	Marcelo
	s1.2 p1	81	jueves, 10 de septiembre de 2015	Retardos. Trabajo de problemas por parejas, resolución y presentación en la pizarra	4.1E03	11:00-13:00	Sofia
		83	miércoles, 9 de septiembre de 2015	Retardos. Trabajo de problemas por parejas, resolución y presentación en la pizarra	4.1E05	17:00-19:00	Sofia
2	s2.1 tc1	M81	martes, 15 de septiembre de 2015	Principios de nivel de aplicación. Servicios ofrecidos por la capa de transporte (TCP/UDP) a la capa de aplicación. Ejemplo de aplicación. DNS. Test de concepto1. temas 1 y 2	4.1E03	11:00-13:00	Marcelo
		M83	martes, 15 de septiembre de 2015	Principios de nivel de aplicación. Servicios ofrecidos por la capa de transporte (TCP/UDP) a la capa de aplicación. Ejemplo de aplicación. DNS. Test de concepto1. temas 1 y 2	4.1E03	17:00-19:00	Marcelo
		M89	Tuesday, 15 September 2015	Application layer principles. Transport layer services (TCP/UDP) offered to the application layer. Application example. DNS. Concept test 1. chapters 1 and 2	1.2G04	9:00-11:00	Marcelo
	s3.1	81	jueves, 17 de septiembre de 2015	Principios de nivel de transporte. Multiplexación y Demultiplexación. UDP	4.1E03	11:00-13:00	Marcelo
		83	miércoles, 16 de septiembre de 2015	Principios de nivel de transporte. Multiplexación y Demultiplexación. UDP	4.1E05	17:00-19:00	Marcelo
3	s3.2	M81	martes, 22 de septiembre de 2015	Transferencias de datos fiables.	4.1E03	11:00-13:00	Marcelo
		M83	martes, 22 de septiembre de 2015	Transferencias de datos fiables.	4.1E03	17:00-19:00	Marcelo
		M89	Tuesday, 22 September 2015	Reliable Data Transfer	1.2G04	9:00-11:00	Marcelo
	p2	81	jueves, 24 de septiembre de 2015	Trabajo de problemas por parejas, resolución y presentación en la pizarra	7.0J04	11:00-13:00	Marcelo
		83	miércoles, 23 de septiembre de 2015	Trabajo de problemas por parejas, resolución y presentación en la pizarra	4.5D01	17:00-19:00	Marcelo
4	s3.4	M81	martes, 22 de septiembre de 2015	TCP	4.1E03	11:00-13:00	Marcelo
		M83	martes, 22 de septiembre de 2015	TCP	4.1E03	17:00-19:00	Marcelo
		M89	Tuesday, 22 September 2015	TCP	1.2G04	9:00-11:00	Marcelo
	pc1 s3.3	81	jueves, 1 de octubre de 2015	Prueba de concepto 1. temas 1 y 2 Applets Capítulo 3	4.1E03	11:00-13:00	Sofia
		83	miércoles, 30 de septiembre de 2015	Prueba de concepto 1. temas 1 y 2 Applets Capítulo 3	4.1E05	17:00-19:00	Sofia
5	s3.5 tc2	89	Friday, 2 October 2015	Proof of concept 1. chapters 1 and 2 Applets Chapter 3	1.2G04	11:00-13:00	Sofia
		M81	martes, 6 de octubre de 2015	Principios de control de congestión. Control de congestión en TCP	4.1E03	11:00-13:00	Marcelo
		M83	martes, 6 de octubre de 2015	Principios de control de congestión. Control de congestión en TCP	4.1E03	17:00-19:00	Marcelo
		M89	Tuesday, 6 October 2015	Congestion control principles. TCP congestion control Concept test 2. chapter 3	1.2G04	9:00-11:00	Marcelo

Classes

- The content explained during the classes are a **subset** of the whole content of the course
 - Students must cover **ALL** the contents as detailed in the table of contents of the course (see Excel file in Aula Global)
 - https://www.it.uc3m.es/fvalera/ro/toc_kurose.xlsx
 - It is assumed that students can access the basic bibliography (Kurose&Ross book)
 - There are several copies in the library
 - Online version (8th edition, access from UC3M or using VPN)
 - https://www.ingebook.com/ib/NPcd/IB_Escritorio_Visualizar?cod_primaria=1000193&libro=12321

Methodology

- Explanation of key concepts
- Exercises
- Labs
- Slides available in Aula Global
 - Per session
- May not be sequential
 - Part in theoretical sessions, other parts in exercises sessions, labs or as personal homework

Exercises classes

- Selected exercises from the book will be published in Aula Global so students can prepare them before the class
- Exercises will be done individually
- In the class, the teacher will select two exercises so students can prepare each one during 15 minutes without material
 - Data from the book may be changed
- After 15 min. The teacher will discuss in the blackboard the exercises

Concept tests and proofs of concept

- Concept tests (TC1-TC2)
 - Short tests to be done in class using the online platform
 - The teacher will explain the answers in class
- Proof of concept (PC1-PC4)
 - Several exercises to be solved in class using the online platform
 - There will be 4 of them along the course
 - PC1 introduction and application layer, PC2 transport layer, PC3 network layer, PC4 link layer
 - The teacher will explain the answers in class

Final exam

- No final exam: 100% continuous evaluation
 - only for those that do not have any continuous evaluation grade

Labs

- Network design lab exercise
 - Will be done as homework, and must be delivered using Aula Global before we solve in class
- Lab practices
 - Routers. Will be done in class. Evaluated in class
 - Will be done individually

Assessment (100% continuous evaluation)

- Exercise evaluation (50 points):
 - 4 knowledge tests, on-site (**PC1, PC2, PC3, PC4**): 10 points PC1 and PC4, 15 points PC2 and PC3. Every test will last a maximum of 1.5 hours and will be done in class.
- Theoretical evaluation (10 points):
 - 2 face-to-face concept tests (**TC1, TC2**): 5 points each. Each test will have a maximum duration of 15 minutes. They will be held during class hours.
- Labs evaluation (40 points):
 - IP addressing design (**Ld**, it will be done as homework and delivered using the corresponding activity in the online platform Aula Global): 5 points.
 - Lab practice, on-site (routers **Lr**): 35 points.
- SEE COURSE EVALUATION RULES FILE (AULA GLOBAL)

Optional test (t1-t4)

- They will be done as homework using the online platform. The calendar of these is indicated in the planning file of the subject.
- Each test will last an estimated 5-10 minutes
- The score obtained in these tests will only be used in the event that the final grade is between 4.5 and 5 points, 5.5 and 6 points, between 6.5 and 7 points, between 7.5 and 8 points, between 8.5 and 9 points and above 9 points
- If all the tests have been passed and at least 25% of the points have been obtained in each one of the knowledge tests (PC), an increase in the grade would be considered to allow the grade of 5, 6, 7, 8, 9 points or honors to be achieved as appropriate